DE–7811 **11**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Describe the various generation of computers.

   (b) "Computer – All purpose machine" – Discuss.

2. (a) Explain the four types of application software.

   (b) Discuss about word processing software features.

3. (a) Explain about internet web browsers.

   (b) Explain about intellectual property rights.

4. (a) Describe the telephone related communication services.

   (b) Discuss about electronic data interchange.

5. (a) Write short notes on cyber ethics.

   (b) Write short notes on video conferencing.

6. (a) Describe the features of DBMS.

   (b) What are the types of database organisation? Explain them briefly.

7. (a) Explain briefly about multimedia.

   (b) Explain about internet programming.

8. (a) Explain the various phases of system analysis and design.

   (b) Discuss the issues in information security.

DE–7812 **12**

DISTANCE EDUCATION

PGDCA (S) EXAMINATION, MAY 2014.

First Semester

DIGITAL COMPUTER ORGANISATION

(2005 – 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Convert the following.

   (i) 163.875_{10} to binary

   (ii) 110101.101010_2 to octal. (10)

   (b) Subtract the following binary numbers 1100.10 – 111.01. (10)

2. (a) Describe in detail about the operation of a multiplexer. (10)

   (b) Obtain the minimal 'SOP' expression for $\sum m(2,3,5,7,9,11,12,13,14,15)$ and implement it in NOR logic. (10)

3. (a) Write a note on integer representation. (10)

   (b) Explain in detail about the parallel binary adder. (10)

4. (a) Explain addition and subtraction in a parallel arithmetic element. (10)

   (b) Describe the principle of operation of full adder. (10)

5. (a) Write a note on BCD counters. (10)

   (b) Explain about magnetic disk memory in detail. (10)

6. (a) Write a note on Gated Flip-Flop. (10)

   (b) Explain about Linear select memory organization. (10)

7. (a) Write a note on execution cycle. (10)

   (b) Explain about DMA in detail. (10)

8. Write short notes on

   (a) Programmed I/O. (10)

   (b) Interrupts. (10)

DE–7813 **13**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain any three looping statements in C with examples. (10)

   (b) Explain the terms : Keywords, Constants and Variables with examples. (10)

2. (a) Write a C program to sum all the odd numbers from 100 to 300. (10)

   (b) Explain the method of reading and writing strings in C. (10)

3. (a) Write a recursive function to find factorial of a given number. (10)

   (b) Explain the method of declaring and using two dimensional array with an example. (10)

4. (a) What is C preprocessor? Explain with an example. (10)

   (b) What is Pointer? Explain its use with an example. (10)

5. (a) Explain the functions used for writing data into the file. (10)

(b) Write a C program to find sum of n given numbers. (10)

6. (a) Explain any two operations on stack. (10)

(b) Discuss in detail about double-ended queue with an example. (10)

7. (a) Explain the method of creating a linked list. (10)

(b) Explain the difference between tree and binary tree with examples. (10)

8. (a) Explain any one of the applications of tree. (10)

(b) Explain the method of representing a list as binary tree. (10)

DE–7814 **14**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014

First Semester

OFFICE AUTOMATION

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the procedure of

   (i) working with windows elements

   (ii) working with multiple windows

   (b) Explain the Office Shortcut Bar.

2. (a) How do tables are handled in MS-word? Explain.

   (b) Explain the printing mailing labels.

3. (a) Write short notes on the Mail merge.

   (b) How to enter, modify, insert, delete and copy data cell in the excel sheet? Explain.

4. (a) How to create ,search and sort a database in Excel? Explain.

   (b) Explain the following features :

      (i) Find and replace data

      (ii) Number formatting

5. (a) How to create, edit and run the power point presentation? Explain.

   (b) How to create graph charts and implemented in the power point presentation? Explain.

6. (a) Give suitable operation sequence to do the following :

      (i) Insert a picture in the slide

      (ii) Insert sound in the slide

      (iii) Animated the pictures.

   (b) Explain the various data types in MS access.

7. (a) State the procedure of creating, saving and editing the forms? Explain.

   (b) Narrate the procedure of creating and printing the reports.

8. (a) Explain the usage of the clipboard.

   (b) Write the procedure of creating and saving tables in MS-Access? Explain.

DE–7815 **21**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write down the benefits of OOP. (10)

   (b) Explain the various data types available in C++. (10)

2. (a) Discuss about conditional statements with an example. (10)

   (b) Explain about structures with an example. (10)

3. (a) What is function? Explain with an example. (10)

   (b) Discuss about macros and inline functions with an example. (10)

4. (a) Write a C++ program to find maximum and minimum values in an array. (10)

   (b) Explain about multidimensional array with an example. (10)

5. (a) Explain about operator overloading with an example. (10)

   (b) Discuss about managing console I/O operations in detail. (10)

6. (a) What is virtual function? Explain with an example. (10)

   (b) Discuss about multiple and multilevel inheritance with an example. (10)

7. (a) Discuss about the various file modes in detail. (10)

   (b) Explain about random access file with its operations. (10)

8. Discuss the following with examples:

   (a) Type conversion. (10)

   (b) Error handling during file operations. (10)

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DE–7816 **22**

DISTANCE EDUCATION

PGDCA (Sem) EXAMINATION, MAY 2014.

Second Semester

SYSTEM ANALYSIS AND DESIGN

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the types of systems. (10)

   (b) Discuss the role of system analyst. (10)

2. (a) What is computer based information system? Explain its types. (10)

   (b) Write a note on feasibility study. (10)

3. (a) What is the use of data dictionary? Explain. (10)

   (b) Explain the various fact finding techniques in system analysis. (10)

4. (a) Explain the process and stages of system design. (10)

   (b) Discuss the requirements of form design. (10)

5. (a) Explain the different types of databases. (10)

   (b) What is the need for system testing? Explain the testing techniques. (10)

6. (a) Explain the various steps in system development and software design. (10)

   (b) Write a note on cost benefit analysis. (10)

7. (a) Write a note on system evaluation and implementation. (10)

   (b) Explain the major phases in hardware and software selection. (10)

8. What are the phases in the system development life cycle? Explain in detail. (20)

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DE–7817 **23**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

1. (a) How to develop a VB application?

   (b) Give the importance of Tool Box.

2. (a) Mention the properties of window.

   (b) Explain the concept of Event procedures.

3. (a) List the properties of Text Box.

   (b) How to trigger a Form?

4. (a) Explain the concept of Printer Object.

   (b) With an example, discuss the execution of IF then Else statements.

5. (a) With examples discuss on date functions.

   (b) How function differs from a procedure?

6. (a) Give the uses of Combo Boxes.

   (b) Explain the concept of Load and Unload statements.

7. (a) What is meant by Error Trapping?

   (b) How to draw the geometric objects in Visual Basic?

8. (a) Compare sequential access with random access in file handling?

   (b) Mention the usage of Clipboard.

DE–7818 **24**

DISTANCE EDUCATION

P.G.D.C.A.(S) EXAMINATION, MAY 2014.

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Give a neat description on any four input devices with neat diagrams. (20)

2. (a) Explain the line drawing algorithm with program segment. (12)

   (b) Write down the midpoint circle generation algorithm. (8)

3. List and explain the basic transformation principles of a two dimensional object. Explain with matrix representations. (20)

4. Describe the Cohen-Sutherland line clipping algorithm with program segment. (20)

5. Write notes on the following principles for a three dimensional object :

   (a) Scaling (7)

   (b) Translation (6)

   (c) Mirror reflection. (7)

6. Give a neat description on three dimensional viewing pipeline and coordinates concepts. (20)

7. Write notes on :

   (a) Hidden surface "depth sorting" method (10)

   (b) Scan line method. (10)

8. Discuss the user interface design concepts "User's model" and "Command language". (20)

DE–8781 **15**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014.

First Semester

COMPUTER LAB – I DATA STRUCTURES USING C

(2005 to 2010 Academic year)

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by LOT system

(Programs must be written in C)

1. (a) Write a program to implement stack operations using arrays.

   (b) Write a program to demonstrate various string manipulations.

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2. (a) Write a program to insert and delete an element in queue using arrays.

   (b) Write a program to multiply two matrices provide the compatiality check of matrices for multiplication.

3. (a) Write a program to find the roots of quadratic equation $ax^2 + bx + c = 0$.

(b) Write a program to arrange the given 'n' positive integers in to ascending order.

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4. (a) Write a program to count the number of lines and number of characters in a file.

(b) Write a program to calculate factorial of a positive integer.

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5. (a) Write a program to write the given integer into words.

(b) Write a program using structure for mark list of a class.

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6. (a) Write a C program to insert and delete operations on linked list.

(b) Write a program to sort 'n' numbers using selection sort.

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7. (a) Write a program to evaluate the given arithmetic expression using stack.

(b) Write a program to check the given word as palindrome or not (eg : LIRIL is a palindrome and HAMAM is not a palindrome)

8. (a) Write a menu driven program for matrix manipulations.

(b) Write a menu driven program to find an element using linear and binary search methods.

DE–8782 **16**

DISTANCE EDUCATION

P.G.D.C.A. (Sem) EXAMINATION, MAY 2014.

First Semester

LAB II — MS OFFICE AND INTERNET

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging / Execution	:	35
Result	:	10
Total	:	100

ONE question should be selected by the Examiner and given to the candidates by lot system.

1. (a) Create a worksheet is MS–Excel with Employee number, Employee name, designation Basic pay, Hra, Da, Lic, Pf, Gross pay, Netpay.

   (i) Type data for empno, empname, designation, Basic pay and Lic, Pf.

   (ii) Calculate Hra = 20% of basic
   Da = 30% of basic
   Gross pay = Basic +Hra+Da
   Netpay = Gross pay – (Lic+pf)

   (iii) Draw bar chart between empname and net pay.

   (b) Create HTML page to display 10 courses and their highlighted features in a colorful format.

2. (a) Prepare a news report using two columns, insert a picture in the first column and make the text flow around it.

   (b) Create HTML page that plays video and audio files.

---------- Cut here ----------

3. (a) Create a worsheet with the following columns

   Sales man number, Sales man Name, City, product name, sales amount

   Add three records for 5 different salesman who have carried out sales of different products in different cities, Find the following.

   (i) Citywise total sales

   (ii) Salesman wise total sales

   (iii) Product wise total sales

   (iv) Grand total sales

   (b) Create HTML page that divides the screen into three frames (one vertical and two horizontal of equal size) display three different files in them with scrolling titles relevant to their contents.

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4. (a) Create STUDENT table with ragno, internal and external marks for five subject, create another table SUBJECT with subject code and subject title for five subjects. Create a report to print marksheets for all the students using the relevant fields and suitable headings.

   (b) Create HTML page that receives employee name, age, street, city, pin data. Receive male/Female and married/ hobbies through check boxes.

5. (a) Prepare the first page of your practical record note with picture insertion and alignment.

(b) Create HTML page that divides the screen space into two equal frames either vertically or horizontally. Display the picture of a animal in one frame and display the details about the animal in another frame.

---------- Cut here ----------

6. (a) Create two tables, SALES1, SALES2 and join them to produce a third table SALES3.

(b) Create a HTML page to display the details of the course. Link to another page when user clicks over the course. The linked page should display the features of the course clicked.

---------- Cut here ----------

7. (a) Create two worksheets containing day to day house hold expenses for the months January and February 2011. Find the total expenses for two months. Also find the maximum and minimum expense amount in both the months.

(b) Create a greeting card in HTML with attractive pictures, scrolling text, and colorful background. Audio can also be attached.

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8. (a) Create a new document and type the following mathematical expression.

$$\int_{0}^{\infty} xdy / dx \sum_{i=1}^{100} ax_i + b + |A| \pm \sqrt[3]{ax + b\lambda}$$

(b) Create a HTML program to create a website about you university explaining the features in different pages and give corresponding link to activate it.

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DE–8783 **25**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014.

Second Semester

COMPUTER LAB III – GRAPHICS USING C++

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

ONE question should be selected by the examiner and given to the candidates by lot system.

Each questions carries 100 marks.

Break-up of marks :
Record Note Book : 5
Algorithm : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. Develop a C++ program to manipulate the graphical object using shading technique.

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2. Write a C++ program to draw a circle within a square in such a way that the diameter of the circle is equal to side of the square.

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3. Develop a C++ program to draw a right- angled triangle and rotate it in clockwise by 90°.

4. Write a C++ program to draw a ball and perform animation on it.

---------- Cut here ----------

5. Develop a C++ program to draw a circle with four equal quadrants and fill each quadrant with unique colour.

---------- Cut here ----------

6. Write a C++ program to draw non-linear objects arcs and curves.

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7. Develop a C++ program to draw a straight line using bresenham's line drawing technique.

---------- Cut here ----------

8. Develop a C++ program to implement simple DDA technique for line drawing.

---------- Cut here ----------

9. Write a C++ program to draw non-linear object "ellipse".

---------- Cut here ----------

10. Develop a C++ program to show graphical transformation on three dimensional object.

DE–8784 **26**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2014.

Second Semester

COMPUTER LAB IV — VISUAL BASIC

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart/Syntax	:	15
Writing Program/Coding	:	35
Debugging and Execution	:	30
Result/Output	:	10
Viva voice	:	5
Total	:	100

ONE question should be selected by the examiner and given to the candidate by lot system.

1. Write a VB project that receives a foreign currency value selected from a list box and converts it into equivalent Indian rupees (e.g US Dollar 42.45, sterling 71.30, D. mark 25.52, SW Franc 31.58, Saudi Riyal 11.40, French Fran 7.60).

2. Write a VB project to create a screen saver that displays a list of pictures and 1 second pause in between successive pictures.

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3. Write a VB project using Built in AX control (Rich Text Box), develop the windows NOTE PAD like Editor with File and Edit Menus and also display the floating menu whenever necessary.

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4. Write a VB project for a Blood Bank that maintains a list of donors with address and their blood group. Provide the following reporting features :

   (a) Search and display the address of a particular donor, given the name in a text box.

   (b) Display all the donors (using data report)

      (i) In age group 20 – 30

      (ii) In particular city

      (iii) With particular blood group

      (iv) Male donors with particular blood group

      (v) Female donors with particular blood group.

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5. Write a VB project using ActiveX control to create a Text box that accepts only numeric value. Provide the following properties for the text box :

   Back color, fore color and text.